



Department of Computer Science and Engineering

Academic Year 2022– 2023(Even Semester)

Degree, Semester & Branch: VI Semester B.E. CSE

Course Code & Title: CS8603 & Distributed Systems

Name of the Faculty member (s): Dr.V. Anusuya, ASCP/CSE

Innovative Practice Description

- **Unit / Topic:** Unit 3 / Algorithms for the single resource model
- **Course Outcome:** The student will be able to Comprehend the idea of the Mutual Exclusion and Deadlock detection algorithms in distributed systems
- **Topic Learning Outcome:** TLO7.
- **Activity Chosen:** Flipped Class Room
- **Justification:**
 - The concept of algorithm for the single resource model video was posted into the student's group, to identify the learning level of students about the single resource model activity was given. It is an important topic the question was asked in most of the university questions.
- **Time Allotted for the Activity:** 20 Minutes
- **Details of the Implementation:**
 - The material for the activity were given to the students one week before through the mail and canvas. The students were asked to watch the video individually at their home and take the notes according to their understanding level.
 - On the day of activity ask the students to form a group. Each students discussed the topic algorithm for single resource model with their group.
 - Invite everyone back together in class to discuss each group's work. Ask questions and explore their understanding individually.
 - Mr.K.Paranthaman and Ms.M.K.Divya team explained the concept of single resource model as shown in Figure 1 and 2.
- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO8	PO10	PO12	PSO1	PSO2	PSO3
CO3	3	2	1	1	2	2	2	3	3	2

(1 – Low

2 – Moderate

3 – High)



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• PO / PSO mapped:

Innovative practice	PO1	PO2	PO3	PO4	PO8	PO10	PO12	PSO1	PSO2	PSO3
	3	2	1	1	2	2	2	3	3	2
Justification for correlation	The basic knowledge of engineering fundamentals is necessary to get the knowledge in Single resource model	The student should identify the basic knowledge of single resource model for solving computer engineering problems	The student will be Provide the solution for deadlock with the knowledge of single resource model	The student will be able Use the knowledge of single resource model in distributed computing	The student will be able to provide solution for deadlock by following the ethical principle or rules	The student doing the problem by the group of two members	The students improve communication by sharing knowledge in the form of presentation	The concept of algorithm for single resource model is important in mobile application development	The student will be able to provide the distributed solution using the knowledge of single resource model	The student will be able to solve AI and big data analytics problems



- **Images / Screenshot of the practice:**



Figure 1: Discussion on the topic Algorithm for Single Resource Model



Figure 2. Presentation of students M.K. Divya and Mr K. Paranthaman

- **Reflective Critique:**

- ***Feedback of practice from students and other stakeholders:***

- The students are actively participated in the activity.
- The student can able to explore their knowledge on explaining the topic Algorithm for single resource model and its applications.

- This activity helps the students to gain knowledge on the deadlock and solutions for handling the deadlock using the single resource model.
- ***Benefit of the practice:*** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)
 - The students understand well about algorithm for single resource model in distributed systems.
 - The students can able to Interact with the faculty member and their class mates during the class was effective.
 - The fear of students was reduced compare when discussing with their friends.
 - It helps the students to improve the self-learning of a student.
 - With the help of this activity, each student can get clear idea about the use of single resource model in distributed systems.
- ❖ ***Challenges faced in implementation:***
 - Some students are hesitated to come to dias.
 - The activity was planned for 20 Minutes, but it takes 25 minutes because the students hesitate to come dias.

References:

- <https://www.ritrjpm.ac.in/images/computer-science/>
- https://www.ritrjpm.ac.in/images/computer-science/2021-2022/MC_Unit-IV_FlippedClassroom.pdf

Signature of Faculty Member

HOD